

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	VBCUA-01
Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report presents the Voice-Based Concept Understanding Analyser (VBCUA), an AI-enabled educational tool that evaluates a student's spoken concept explanation. It combines speech-to-text transcription, semantic similarity analysis, audio feature extraction, filler word detection, scoring, and PDF report generation to provide clear feedback.

Project Overview	
Objective	To build an AI-based system that analyses a student's spoken explanation, checks similarity with a reference concept, evaluates speech quality, and generates a final performance report.
Scope	The project covers audio upload, transcription, semantic similarity scoring, pause and voice-energy analysis, filler word detection, final scoring, waveform visualization, and PDF report generation through Streamlit.
Problem Statement	
Description	Students often explain concepts verbally, but manual evaluation of conceptual accuracy, fluency, pauses, and filler words is time-consuming and inconsistent.
Impact	This solution supports faster feedback, improved concept clarity, better speaking confidence, and structured performance tracking for students and educators.
Proposed Solution	
Approach	Use Whisper for transcription, Sentence-BERT for semantic comparison, Librosa for audio feature analysis, filler detection, and a weighted scoring engine.
Key Features	- Speech-to-text conversion of uploaded audio explanations. - Semantic similarity score against a reference concept. - Audio duration, pause ratio, and voice-energy analysis. - Filler word detection and communication feedback. - Final score, suggestions, waveform visualization, and downloadable PDF report.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop CPU for local demo; cloud CPU/VPS or Render for online deployment. GPU is optional for faster AI processing.
Memory	RAM specifications	Minimum 8 GB RAM recommended.
Storage	Disk space for audio files, models, waveforms, and reports	At least 5-10 GB free storage.
Software		
Frameworks	Python application framework	Streamlit
Libraries	Additional libraries	openai-whisper, sentence-transformers, transformers, torch, librosa, soundfile, numpy, matplotlib, reportlab, imageio-ffmpeg, pytest
Development Environment	IDE and runtime environment	VS Code / Terminal, Python 3.11, virtual environment
Data		
Data	Source, size, format	User-uploaded audio files such as WAV, MP3, or M4A; reference concept text entered manually. No external dataset or database is required.